



Olevius Continuous Non-invasive Blood Pressure Monitoring

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Introduction

Background

- Cardiovascular Disease (CVD) remains the leading cause of death worldwide^[2]
- 1.39 Billion people^[2] worldwide suffer from Hypertension, a primary risk factor for CVD
- Only 25% keep it under control^[2]
- Effective management requires high volumes of reliable data.

Current Methods

Sphygmomanometer

- Accurate
- Uncomfortable
- Non-continuous

PPG

- Inaccurate
- Comfortable
- On most devices already

Problem

Current blood pressure monitoring solutions force patients to choose between accuracy, comfort, and data volume. A critical gap exists for a device that combines the accuracy of a cuff with the comfort and data volume of PPG.

Customer Requirements

Primary



Accuracy: $< \pm 5$ mmHg MAE



Non-invasive, Comfortable Design



Automatic Rest Detection



Wireless Data Transmission

Secondary



Battery Life: > 2 days



Compact Size: < 15 mm Thick, < 80 g



Production Cost: BOM $< \$100$

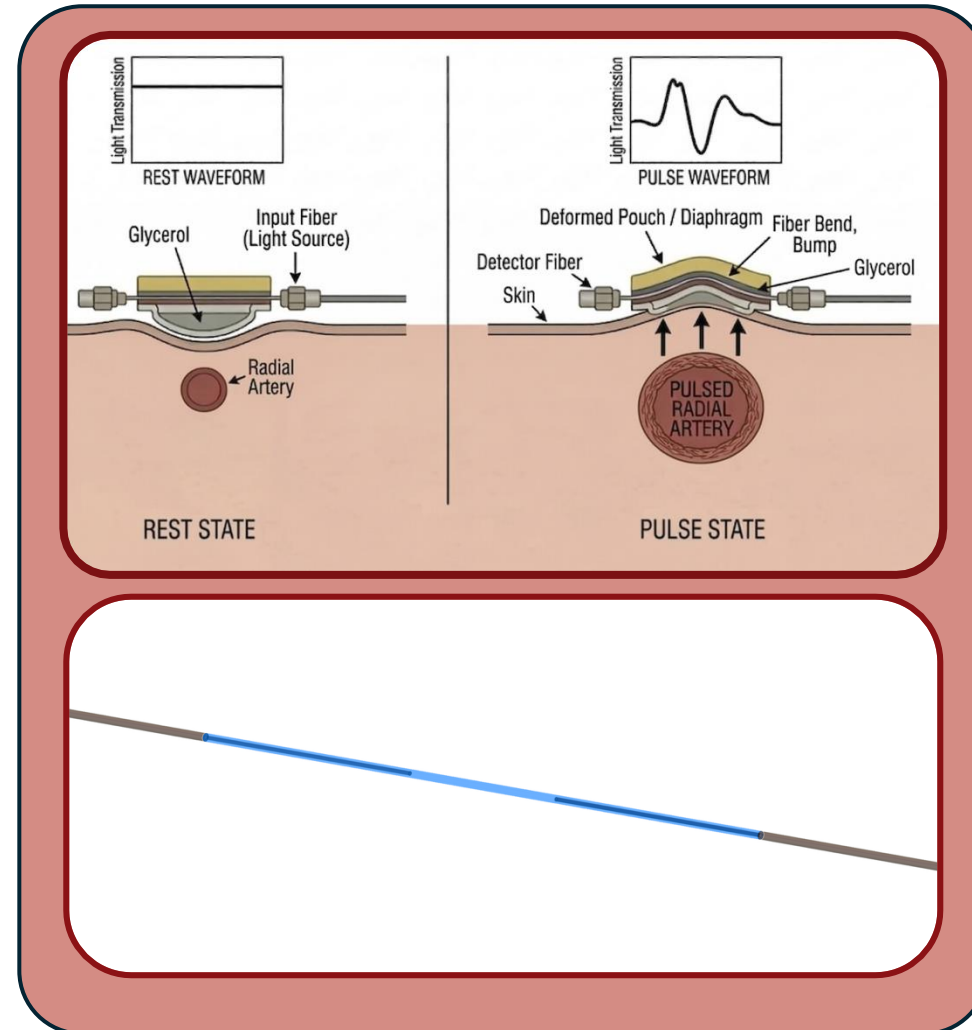
Our Approach

Steady-State

- Light is transmitted across a gap between two fibers

Pulse Event

- Radial artery deforms the sensor face
- Glycerol transmits deformation to the diaphragm
- Fiber joint flexes and misaligns
- Change in transmission creates a pulse waveform



Advantages

Over Cuffs

- Continuous
- Comfortable

Over PPG

- Directly measures BP features
- Accuracy

Trade-Offs

Form Factor

Battery Size

- Larger = longer life
- Takes up a lot of space

PCB Size

- Smaller = more portable
- Very expensive

Sensitivity

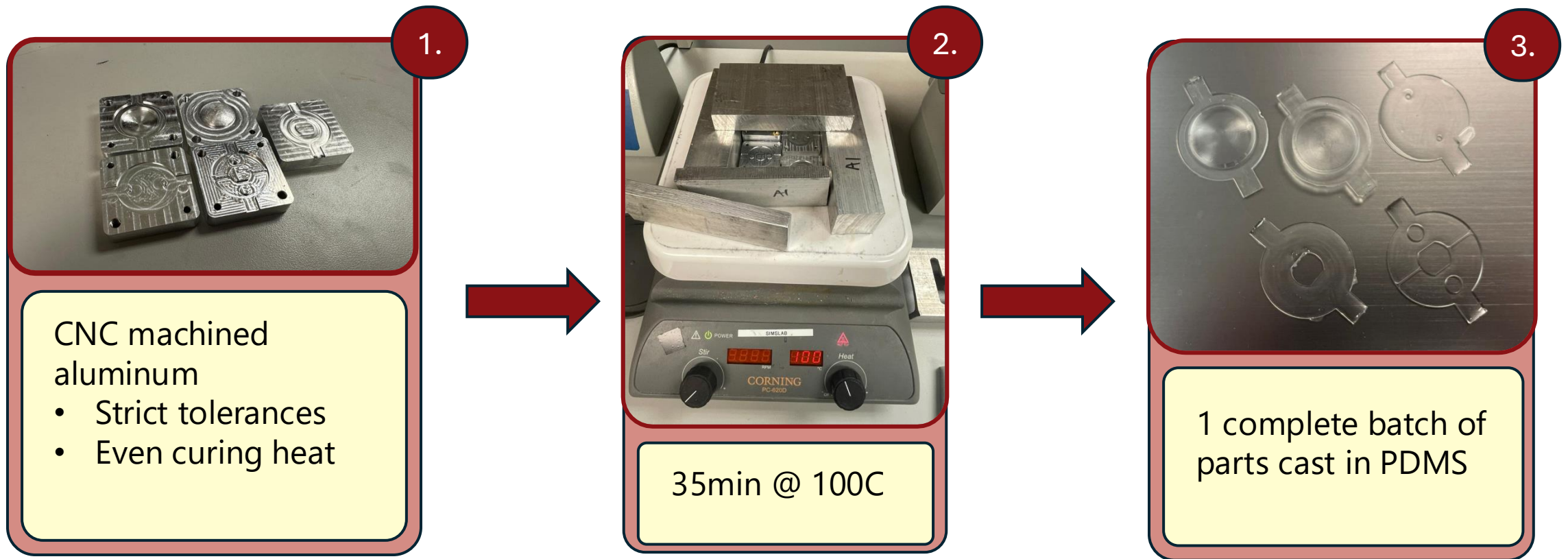
Wall Thickness

- Thinner = sensitive
- Too thin won't appanate artery

Gap Length

- Longer = sensitive
- Loss can be too high

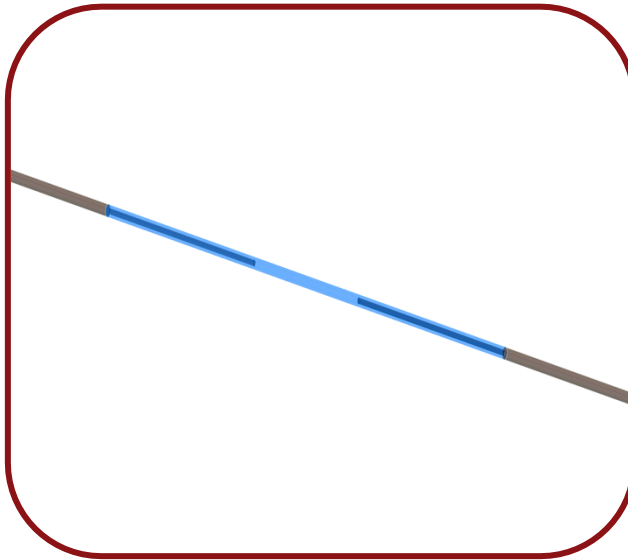
Sensor Construction – Casting



Sensor Construction – Fiber Joint



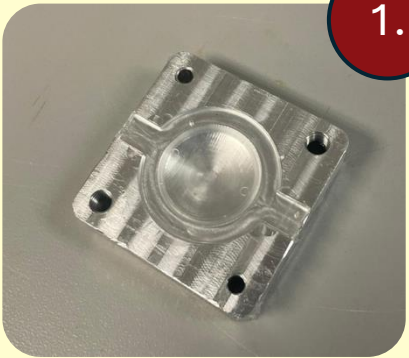
1. Fiber jackets stripped
2. Ends cleaved flat



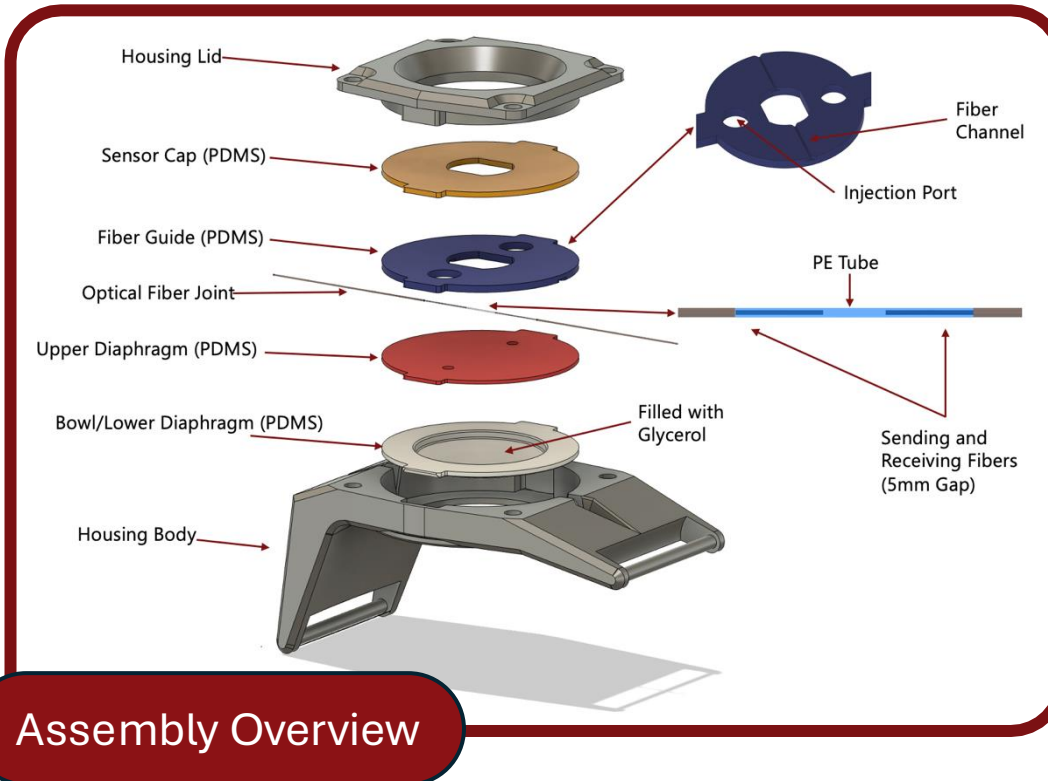
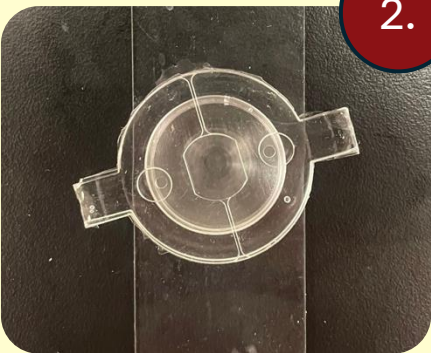
3. Fiber inserted halfway
4. LSE plastic primer
5. CA glue
6. Fiber inserted fully

Sensor Construction – Assembly

1.

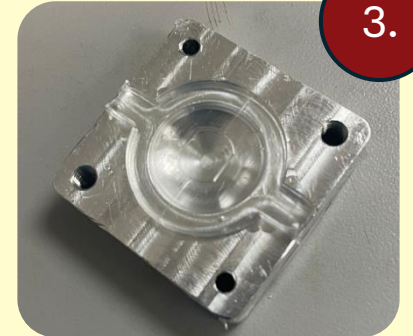


2.



- Parts are plasma-cleaned before bonding
- Hot glycerol injection is most critical step

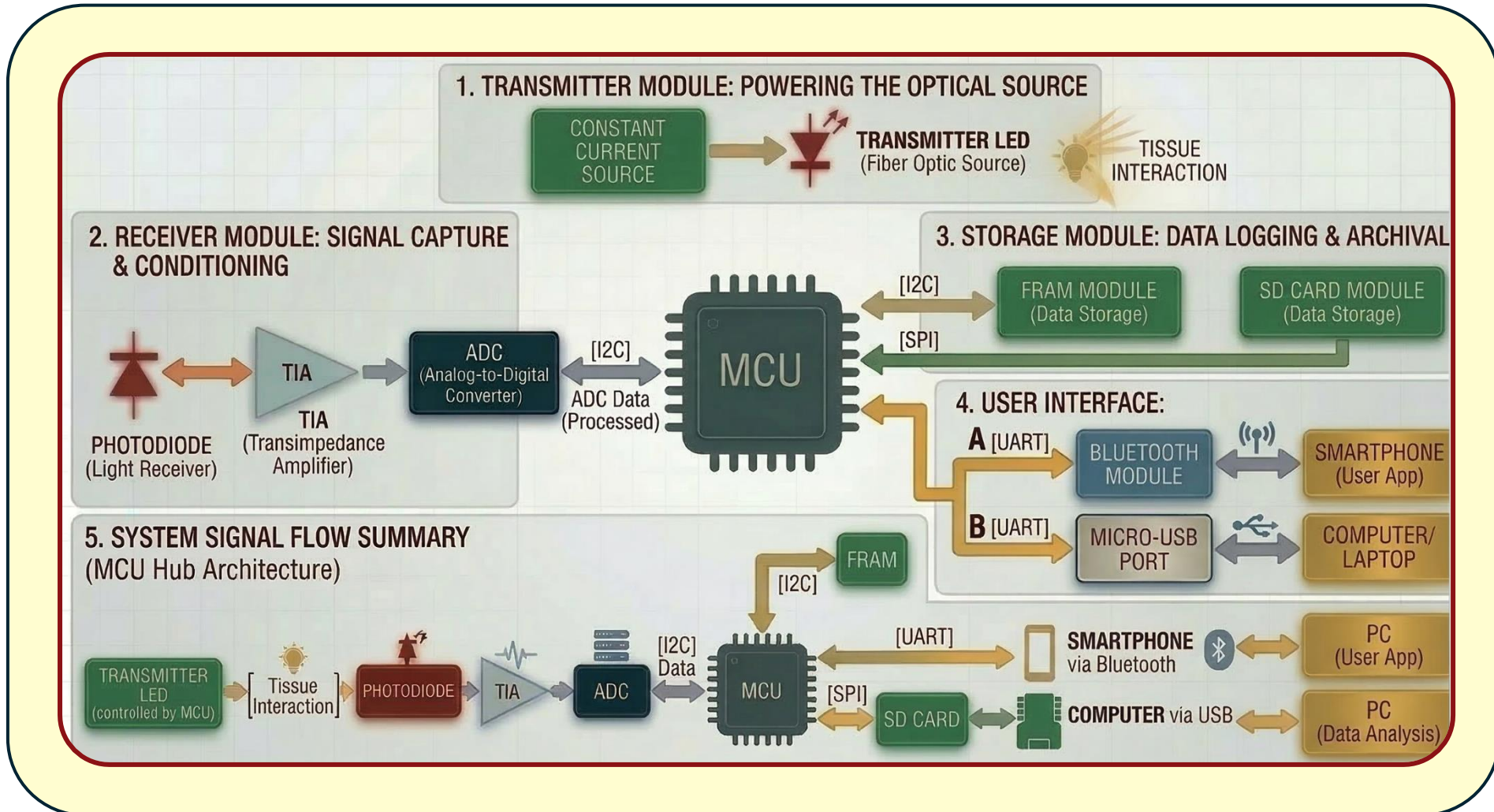
3.



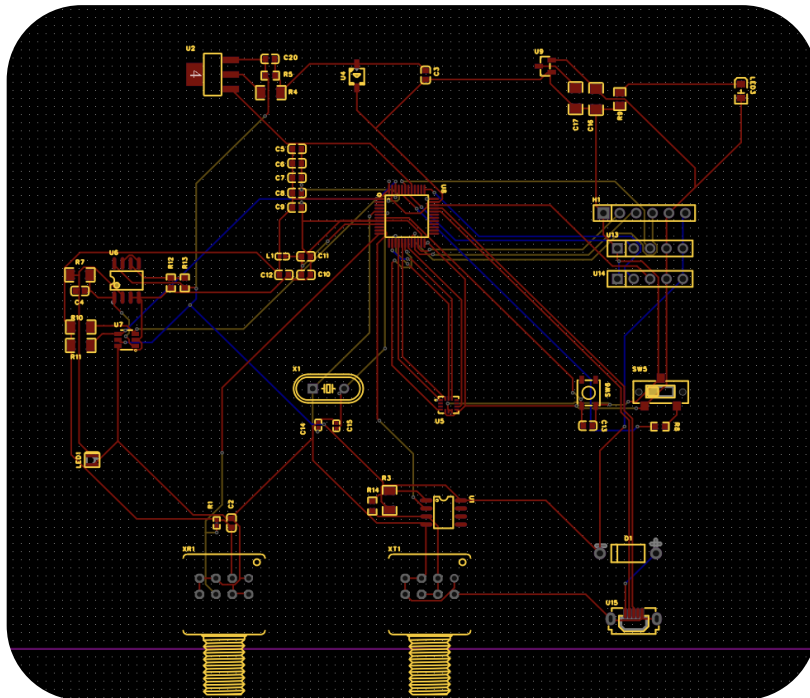
4.



PCB Construction – Block Diagram

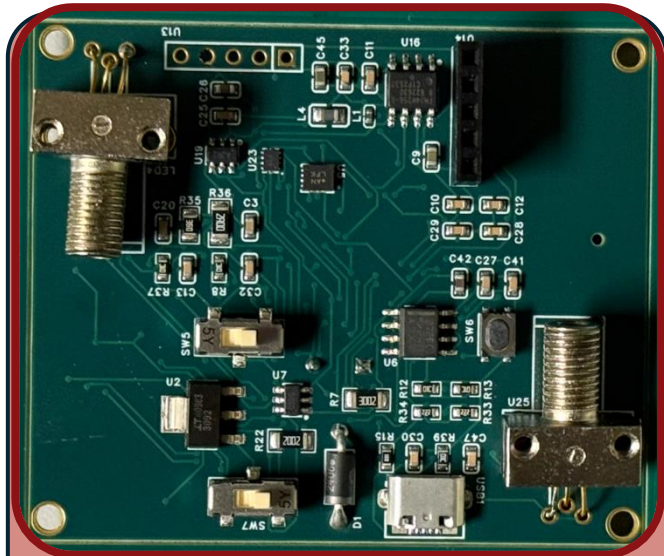


PCB Construction – First Design

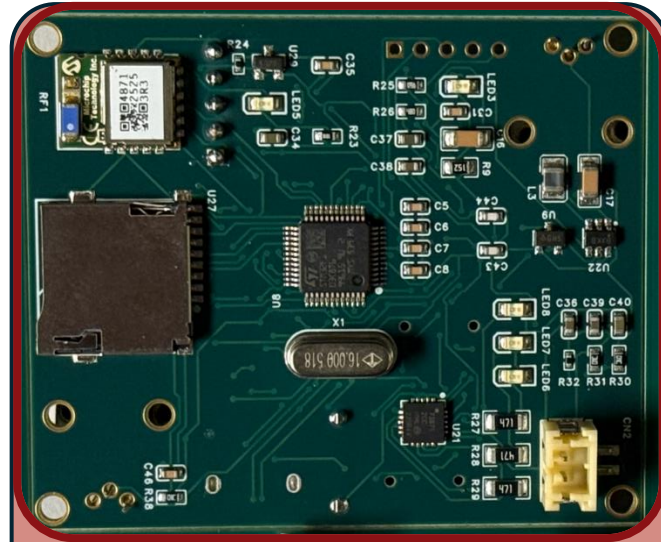


The first PCB design acted as a test board, which was used for tinkering. It consisted of an MCU, ADC, USB, and two separate fiber-optic transmitter and receiver circuits.

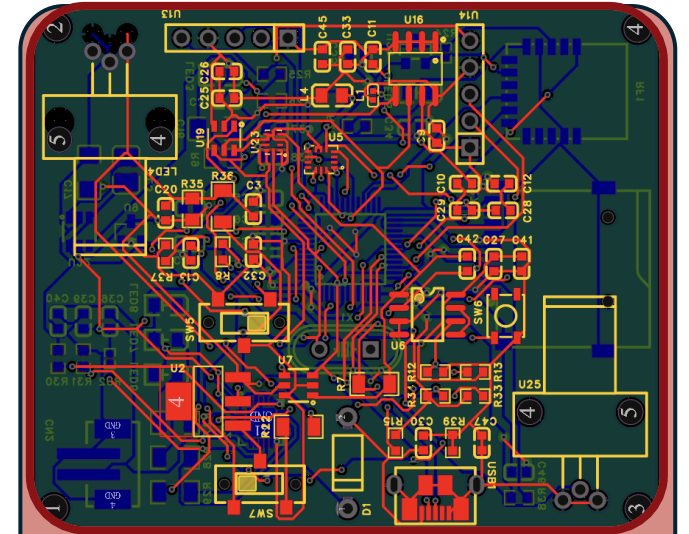
PCB Construction – Second Design



Top View

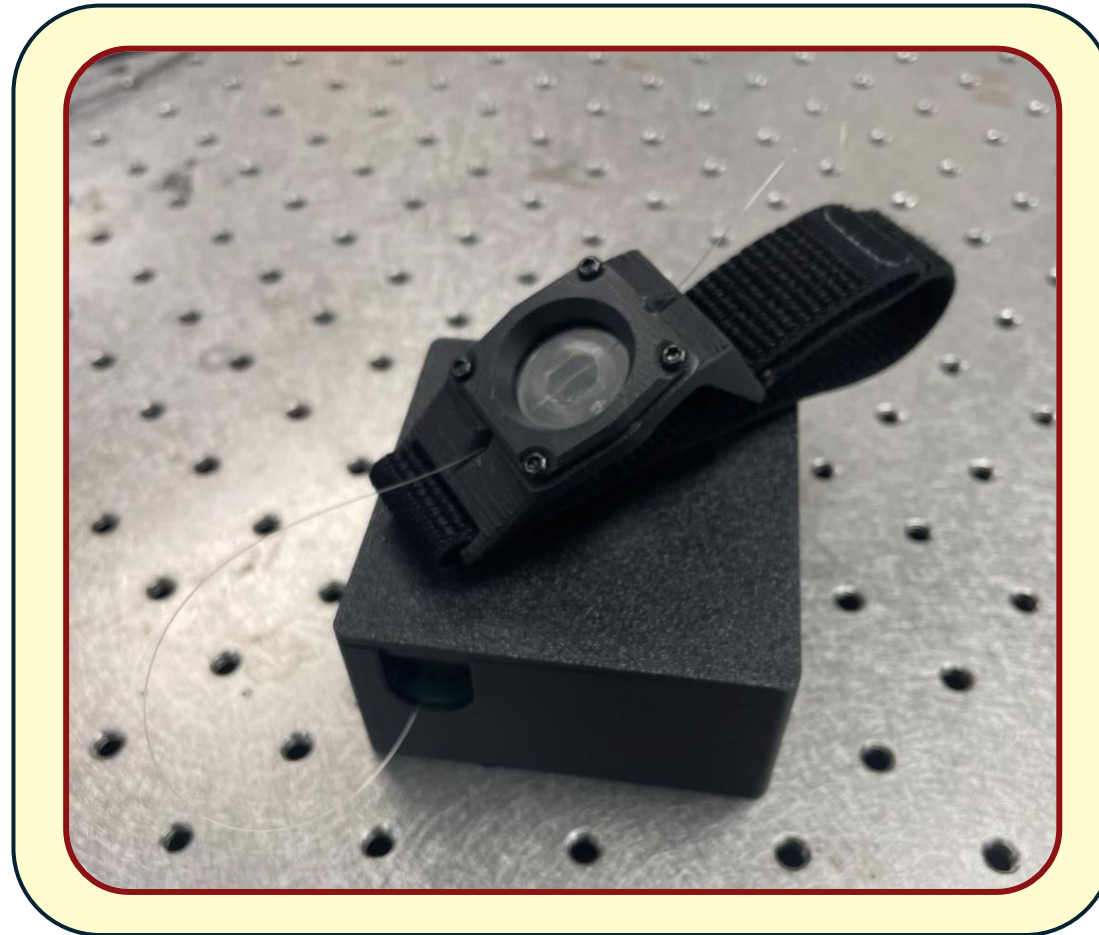


Bottom View



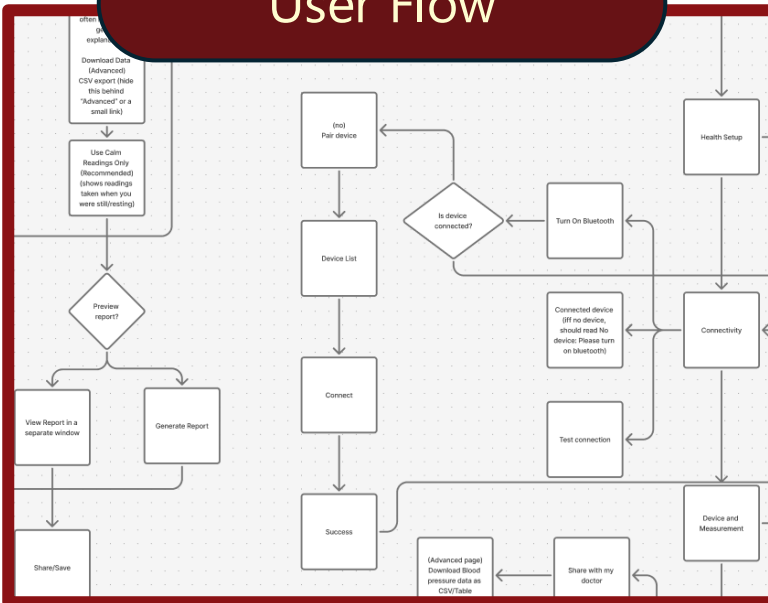
PCB Layout

Final Prototype

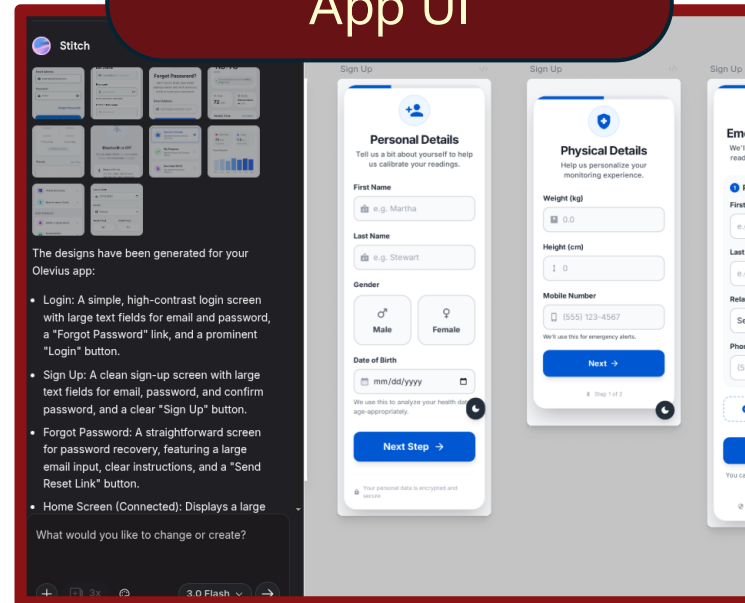


App Design

User Flow



App UI



Development



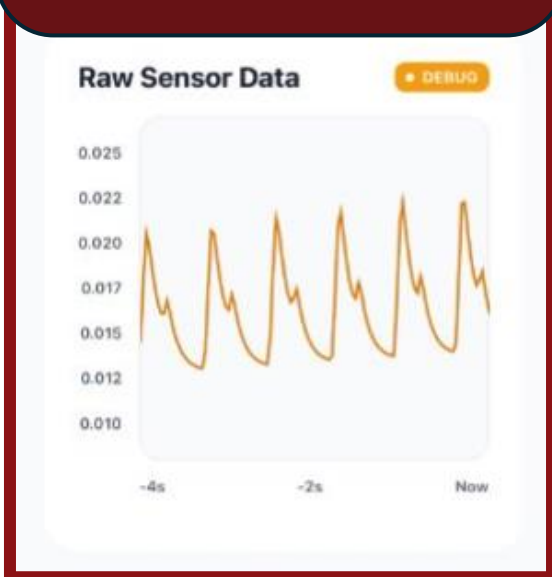
We streamlined authentication, Bluetooth connectivity, and live signal monitoring for a seamless user experience

Generated a Mobile UI using Google Stitch by providing the User Flow diagram as context.

Wrote the frontend code based off the wireframe to develop the mobile interface.

Signal display

PCB Data



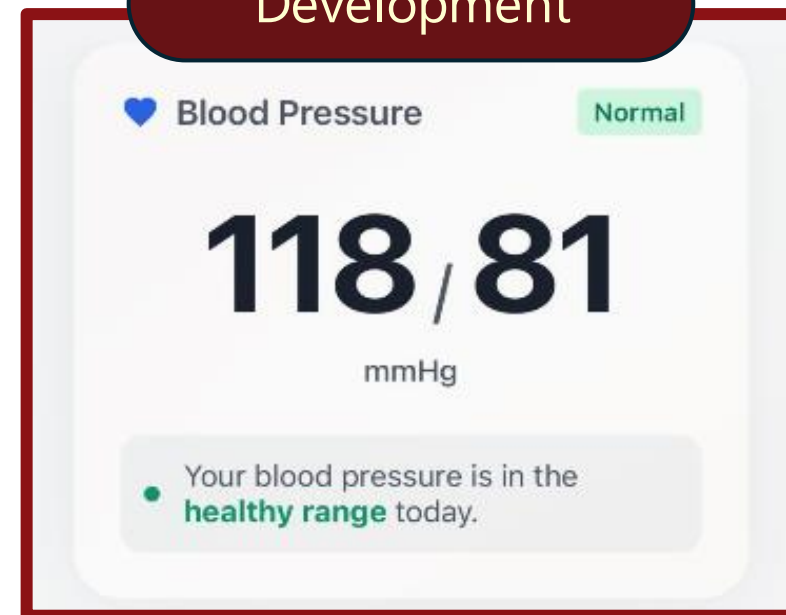
Data is streamed from the PCB via Bluetooth and temporarily stored in a ring buffer to facilitate real-time visualization within the application.

Model Integration

```
async function uploadSampleBuffer(samples) {
  try {
    const res = await fetch(SAMPLE_UPLOAD_ENDPOINT, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ samples, sampleRate })
    });
    console.log('[BLE] Sample upload status: ' + res.status);
  } catch (err) {
    console.warn('[BLE] Sample upload failed: ' + err.message);
  }
}
```

We built an integration to push queued app data to the model endpoint for inference. Future versions will move this process entirely to the user's device.

Development



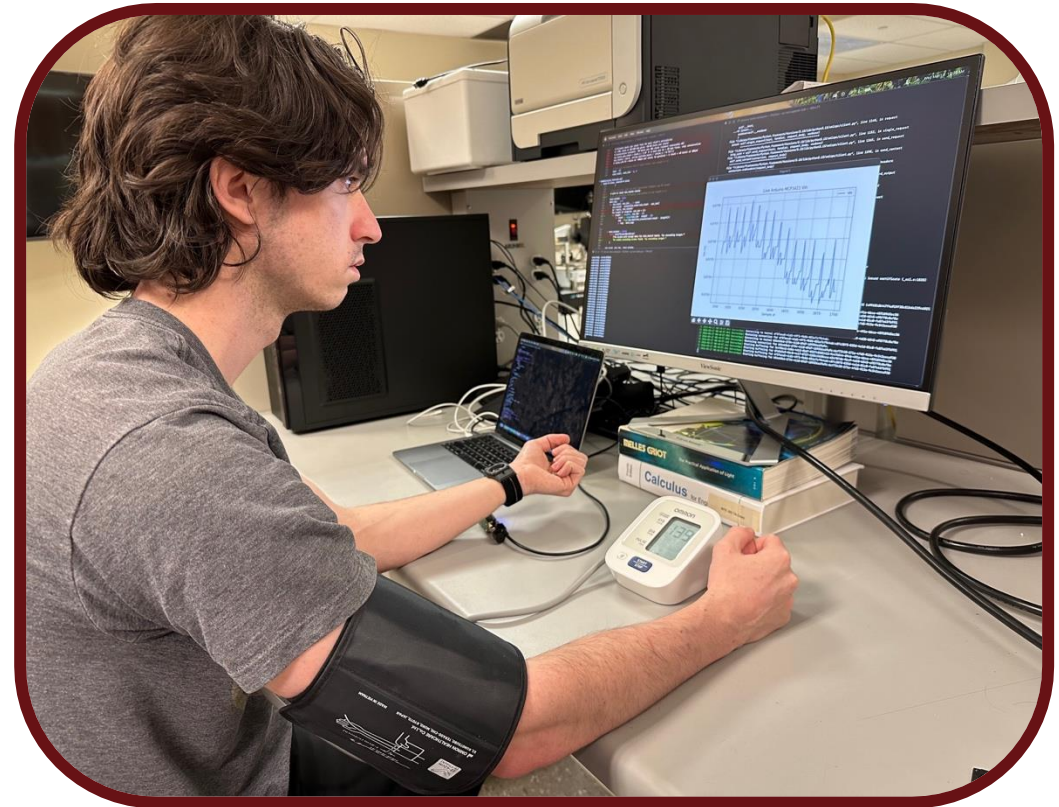
Rendered real-time predictions by buffering returned endpoint data, displaying a continuous sliding window of systolic and diastolic metrics.

Results

Data Collection

The device was worn on the left wrist while reference blood pressure measurements were concurrently obtained on the right arm using a clinically accurate OMRON sphygmomanometer.

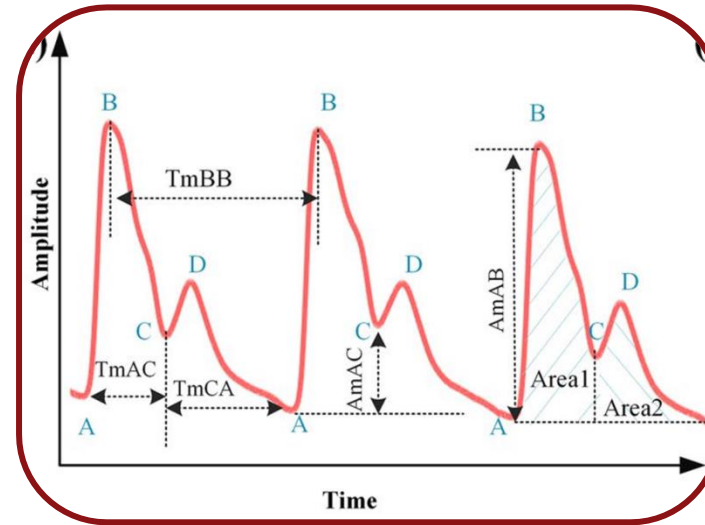
Anonymized data was collected from 14 individuals under varying conditions, including at rest, post-exercise, and at different times of the day.



Results

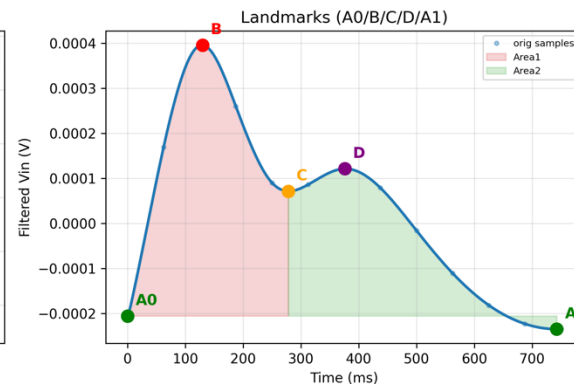
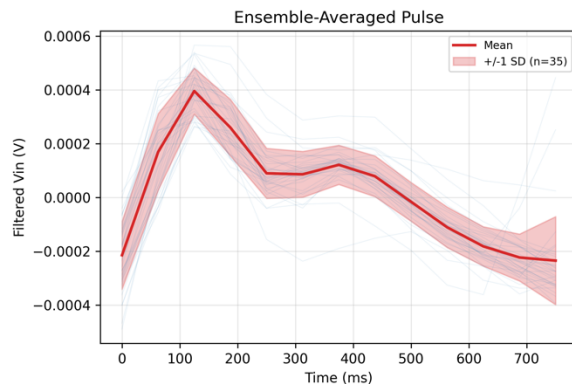
Feature detection

An algorithm adaptively chooses stable, high-quality sampling windows for ensemble averaging. This is done by dividing the recording into 30-second chunks and scoring each chunk using autocorrelation peak height, which directly measures the periodicity (and therefore pulse wave signal quality) of the segment. The contiguous window with the highest average periodicity score is selected, filtering out regions corrupted by sensor settling, hand repositioning, or DC drift.



An ensemble average is constructed by identifying systolic peaks in the bandpass-filtered signal, extracting one-beat windows aligned to each peak, and averaging across ~40 aligned beats.

Points on the ensemble include the beat foot (A0), systolic peak (B), dicrotic notch (C), diastolic peak (D), and next beat foot (A1).



Features

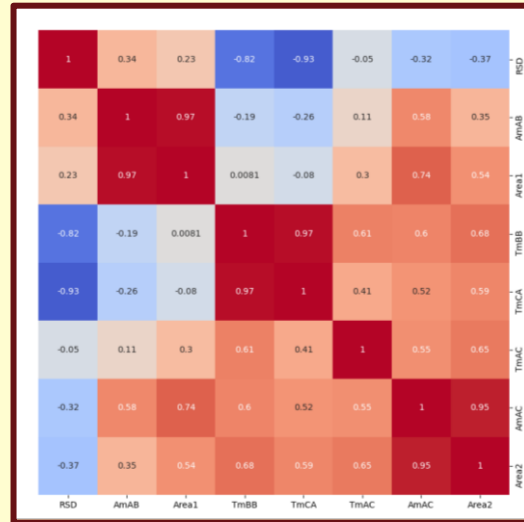
Feature	Value
AmAB	0.000601
AmAC	0.000277
TmBB	0.742481
TmAC	0.278195
TmCA	0.464286
Area1	0.000109
Area2	0.000075
RSD	0.599190
HR (bpm)	73.8
Beats	35
Notch	Yes

From these landmarks we derive systolic amplitude (AmAB), notch amplitude (AmAC), beat-to-beat interval (TmBB), systolic timing (TmAC), diastolic timing (TmCA), systolic area (Area1), diastolic area (Area2), and the reflection symmetry ratio (RSD).

Results

Modeling

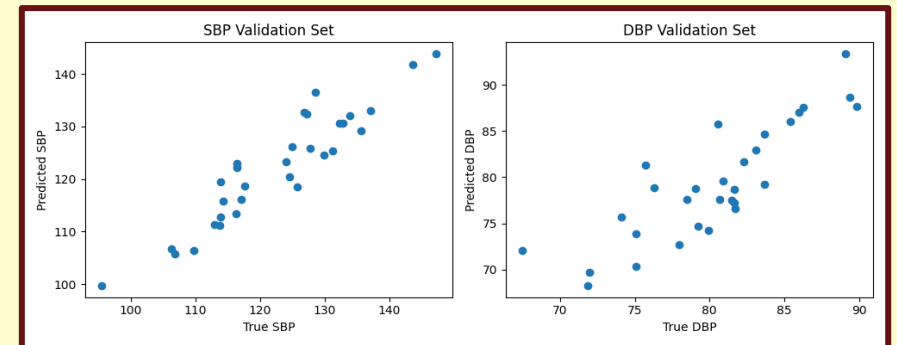
The extracted pulse-wave features from literature were evaluated, and dimensionality reduction techniques were applied to mitigate redundancy and retain the most informative feature set.



The dataset was divided into 70% training and 30% testing subsets, and model performance was assessed using test accuracy, with cross-validation applied to evaluate variability and ensure robustness

Predictivity

The current model achieves mean absolute errors of 3.38 ± 4.00 mmHg (SBP) and 2.87 ± 3.28 mmHg (DBP), within this preliminary dataset.



Summary

Primary Requirements

- Accuracy: Achieved ± 5 mmHg MAE 
- Non-invasive and Comfortable: Yes 
- Wireless Data Transmission: Yes 
- Auto Rest Detection: No, but IMU is there, implementation would take 1 more day 

Secondary Requirements

- Battery Life: 2 Days 
- Compactness: Exceeds size requirement 
- Production BOM Cost: ~2\$/sensor, ~60\$ PCB 

Conclusion

- Olevius proves that accurate and continuous blood pressure monitoring can be achieved comfortably.
- Sensor design is now a validated proof-of-concept.
- Provides a solid foundation for a miniaturized V2 wearable that meets all design criteria.

Acknowledgments

Technical Advice

- Dr. Omar Awad
- Neil Griffett

Consultant

Dr. Patricia Nieva

CNC Machining

- Jason Benninger
- Karl Janzen

References

1. Li, L., Sheng, S., Liu, Y. et al. *Automatic and continuous blood pressure monitoring via an optical-fiber-sensor-assisted smartwatch*. PhotoniX (2023).
2. Mills, K. T., Stefanescu, A., & He, J. *The global epidemiology of hypertension*. Nature Reviews Nephrology (2020).